

The Whole Thing, Explained Simply

What the competition is, what the problem was,
and what we answered

Team Inferno · Round 2 Case Study

Who this is for

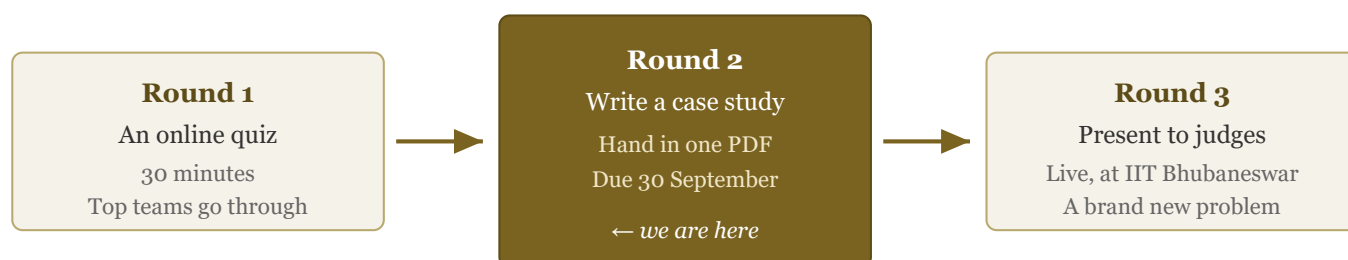
Someone who knows **nothing** about finance, nothing about this competition, and has not read the problem. Everything is explained from zero. Read it slowly. Nothing here assumes you already know anything.

1. What is going on here?

There is a competition. It gave us a made-up business problem. We had to decide what to do and write our answer down. That written answer is what this whole folder is about.

That is genuinely all of it. The rest of this document just fills in the detail.

The competition has three rounds



Round 3 gives you a different problem, so this answer is only used for Round 2.

The rules that matter

Rule	What it means
Teams of 2 to 4	You cannot enter alone. Members can be from different colleges.
Hand in one PDF	Not Word, not a link. One file.
Name the file <i>TeamName_LeaderName</i>	Ours is Inferno_Mahi.pdf
Page 1 = who you are	Every member's name, college, email, phone.
2,000–2,500 words	A hard limit. Too few or too many and you are breaking the rules.
Plus a 500-word summary	A separate short piece at the end.
Deadline 30 Sept, 11:59 PM	Late = rejected. No exceptions are made.

The one that gets people disqualified

Copying. If you copy from anywhere, or reuse something you submitted before, you are out. Everything must be written fresh for this competition, and anything you quote must be credited.

2. Six words you need before anything else

You cannot follow the problem without these. Each one has a tiny example so it sticks.

1. Revenue — money coming IN from sales

You run a juice stall. You sell 100 glasses at ₹10 each. Your **revenue is ₹1,000**. Notice this says nothing about whether you made money.

2. Profit — what is LEFT after you pay for everything

Those 100 glasses cost you ₹900 in oranges, ice and cups. Revenue ₹1,000, costs ₹900, so your **profit is ₹100**.

3. Margin — profit as a percentage of revenue

₹100 profit on ₹1,000 revenue = a **10% margin**. Now imagine you only kept ₹0.20 out of that ₹1,000. That is a **0.02% margin** — and that is the number this whole case turns on. Remember it.

4. Subsidiary — a company owned by another company

Your juice stall company buys an orange farm. The farm is now your **subsidiary**. You own it.

5. Standalone vs consolidated — the single shop, or the whole family

Standalone = just the juice stall's own numbers. **Consolidated** = the juice stall *plus* the orange farm *plus* anything else you own, all added into one big number.

6. Audited — checked by an outsider

Anyone can type a number into a spreadsheet. **Audited** means an independent accounting firm examined the records and signed their name to say the numbers are real. Audited numbers can be trusted. Unaudited numbers are just a claim.

If you only remember one thing from this page

A company can have **enormous revenue and almost no profit** at the same time. That is not automatically suspicious — and understanding why is the whole puzzle.

3. The problem we were given

You are given a role to play. You are an analyst at an investment fund. The fund owns shares in a gold company. Something has gone wrong, and your boss wants your recommendation today.

Who you are	A research analyst at Aarohan Capital , a made-up investment fund that manages ₹4,500 crore of other people's money.
What you own	A small amount of shares in Rajesh Exports , a gold company.
What went wrong	The government's market regulator says it is not satisfied the company's sales figures are real, and has started an investigation.
Your job	Tell your boss what to do with the shares, and explain why.

Now here is the actual problem, using a shop

Forget gold for a minute. Imagine a corner shop.

What the shop SAYS it sold

₹1,00,00,000 (one crore)

The receipts it can actually show you

₹20,000 ← that thin red line is the whole of it

The shop replies: “You have misunderstood. Those receipts only record my profit on each sale, not the full value of what passed through the shop.”

The gap is about 500 times. The shop's explanation might genuinely be true.

So you say: *fine — then show me your sales book, your supplier bills, and your bank statements.*

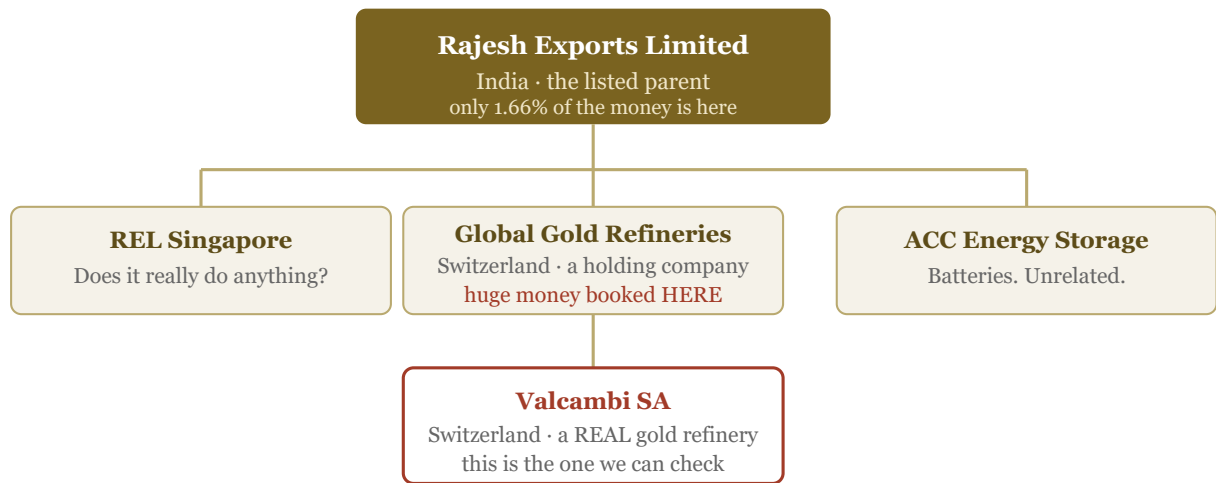
And the shop does not show you.

That is the entire case

A huge claimed number. A tiny checkable number. A reasonable explanation for the gap. And then no paperwork to back the explanation up. Everything from here is just this same shape, with much bigger numbers.

4. The company, and why its shape matters

Rajesh Exports is an Indian company that deals in gold. But it owns other companies in other countries, and that is where nearly all the money supposedly is.



A **holding company** does not make or sell anything. It only owns other companies. So money appearing at one, instead of at the refinery below it, is odd.

The Indian company at the top is tiny. Almost everything sits in the boxes below it.

How lopsided is it?

In the most recent year the group said it took in **₹4,23,099 crore**. Of that, only **₹7,027 crore** came from the Indian company that Indian rules let us inspect. Everything else — **98.34%** — came from the overseas companies.

98.34% — overseas, cannot be inspected

↑ that sliver on the left is the 1.66% we can actually check

So when someone asks “*can we trust this company's numbers?*”, they are really asking about that big gold block — the part nobody outside the company has ever been allowed to look at.

5. The argument, from both sides

Before our answer, you need to understand why this is genuinely hard — and why the company's defence is not stupid.

The company's defence (and it is a good one)

Gold refining works differently from a normal business. A refinery takes in gold, melts it, purifies it, and sells it on. The gold is worth a fortune, but the refinery only keeps a tiny fee.

Think of a money-changer at the airport

Someone hands over ₹1,00,000 to change into dollars. The money-changer keeps ₹200 as a fee. Did the money-changer “earn” ₹1,00,000 or ₹200? Depending on how you write the books, **both answers can be correct**. That is exactly the dispute here.

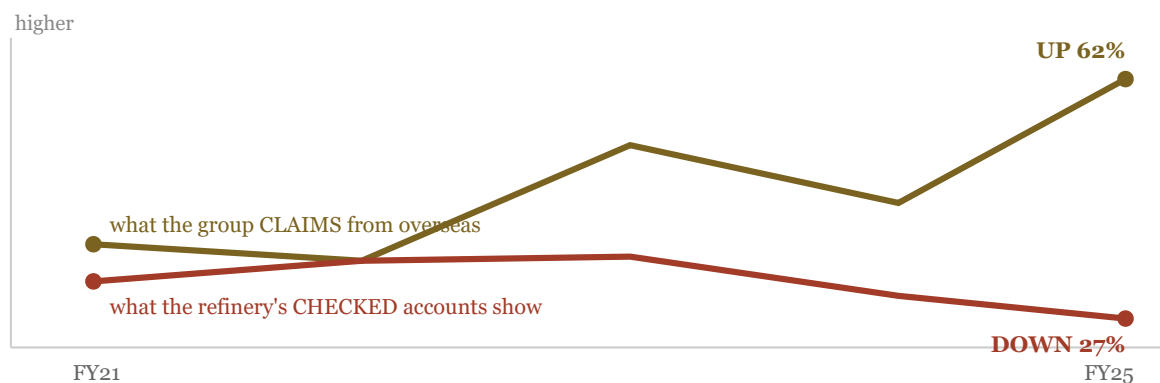
So the company says: the regulator is comparing our *fee* figure with our *total value passing through* figure. Of course one is 500 times the other. Nothing is missing.

This could be true. Our answer says so openly. A good argument has to beat the other side at its strongest, not at its weakest.

Why we did not accept it — three reasons

Reason 1 — the two numbers move in opposite directions

If the big number really comes from the refinery's work, then when the refinery gets busier, both numbers should rise together. They do not.



One line climbs, the other sinks. The money-changer explanation cannot produce that.

Here is the key. A refinery is paid a fee **per ounce** it handles, so its fee tracks *how much* gold passes through, not what gold is worth. The top line's 62% rise roughly matches the rise in the gold price, so the *amount* of gold was about flat. A flat amount of gold should mean a flat fee. Instead the refinery's checked fee line **fell 27%**. The fee explanation cannot produce that.

The fair objection — and our answer

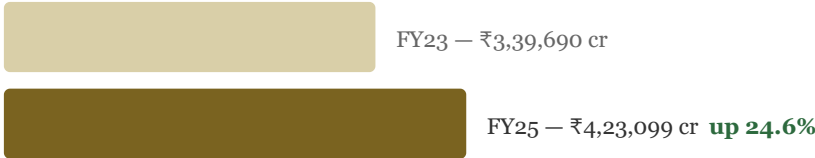
Someone sharp will say: *the top line covers ALL the overseas companies, the bottom line covers only Valcambi. Maybe the work just moved to a different company in the group.* That is a fair point and we say so in the memo.

But it costs them. If the work moved, it moved to companies that **have no gold refinery at all** — which makes things worse, not better. And it would be settled in one page: a simple breakdown of which company earned what. That breakdown has never been produced.

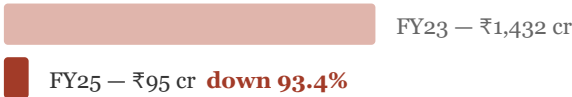
Reason 2 — they sold more but earned almost nothing

When the gold price rises, a gold trader's profit *percentage* shrinks — the same fee is a smaller slice of a bigger number. But its profit *in rupees* should hold up. Here, profit in rupees nearly vanished.

Sales (revenue)



Profit



Sales grew by a quarter. Profit fell by almost everything.

There is a sharper version. Compare the part we *can* inspect with the part we *cannot*:

In the most recent year	Revenue	Profit	Margin
The Indian company — audited, inspectable	₹7,027 cr	₹24 cr	0.34%
The overseas companies — nobody can inspect	₹4,16,072 cr	₹71 cr	0.017%

The visible part earns **twenty times better** than the invisible part. The bit we can check behaves like a normal business. The bit we cannot check does not.

The fair objections — and our answers

“Jewellery earns more than refining, so of course the parent does better.” But in FY22 the overseas companies actually earned **more** than the parent (0.42% against 0.37%). So that is not the reason.

“Maybe they lost money on gold they were holding, or took a one-off write-off.” Possible — but none has been disclosed, and a one-off charge would not repeat across **two years** of falling rupee profit.

Reason 3 — gold that heavy cannot move quietly

This is our strongest point and it works backwards from what you would expect. Read it twice.



600 tonnes is about 12% of all the gold supplied worldwide in a year (World Gold Council).

You cannot move that much metal without leaving a paper trail.

Why this is clever, in one sentence

We do **not** say the amount is too big to be real — it isn't, and saying so would be easy to attack. We say the opposite: if it *is* real, then mountains of paperwork must exist, much of it held by other companies in other countries. So the missing paperwork is *more* damning, not less.

The fair objection — and our answer

“Those records are confidential, they name our customers.” True, but it does not work. A reconciliation statement and an auditor's sign-off name **nobody at all**, and the key reconciliation belongs to the Indian parent company, so Swiss secrecy rules do not cover it. Confidentiality explains why records are not *published*. It does not explain why they were not *handed over*.

6. What we decided

There were four allowed answers. We had to pick exactly one.



Why D and not C — the difference matters

C sells because the stock looks unsuitable. D sells too — but it also sets a rule:
no buying back at ANY price until the missing records are independently checked.

The reasoning, in four steps



The shares looked “cheap” — but cheap compared to numbers nobody can verify means nothing.

Why not just wait and see?

Because waiting is not free. If the investigation ends badly, the share price collapses and **there is nobody left to buy from you**. You would be stuck holding it. So waiting gives you all of the danger and none of the ability to escape. It only looks like the safe choice.

One thing we are careful NOT to say

We never accuse anyone of fraud. The problem asks you to think like an analyst, not a detective. Our answer is narrower and much harder to argue with: **we cannot measure this, so we should not own it** — and here are the exact six documents that would change our mind.

7. What is finished, and what is not

The thinking and the writing are done. Some admin is not.

Item	State
The memo itself, all six required sections	done
Word count — 2,470 at the strictest count, inside the 2,000–2,500 limit	done
The 500-word summary — 493 words at the strictest count	done (only 7 spare)
One answer ticked (D)	done
File named Inferno_Mahi.pdf	done
Page 1 — team name and leader	done
Page 1 — other members, college, email, phone	NOT DONE
Upload before 30 September, 11:59 PM	not yet

The one job left

Page 1 still has gaps where names and contact details should be. The rules require every member's name, college, email and phone. Right now the page has empty square brackets like **[COLLEGE]** sitting in it.

Submitting it in that state would look careless and could break the rules. Fill those in, rebuild the PDF, and then read page 1 with your own eyes before uploading.

How to finish it, step by step

- 1 Open **submission/memo.md** in any text editor — Notepad works.
- 2 Replace every bit of text in square brackets with the real details.
- 3 If the team has fewer than four people, delete the extra table rows.
- 4 Run this one command:

```
python tools/build.py
```
- 5 It prints the word counts and says **OK** twice. If it does not say OK, something is over the limit — fix it before going on.
- 6 Open the new **Inferno_Mahi.pdf**. Read page 1. Check there are no square brackets left anywhere.

If you remember nothing else from this document

A gold company said it made an enormous amount of money overseas. Almost none of it can be checked by anyone outside the company. Its explanation for the gap is reasonable but unproven, and the paperwork that would prove it has not been handed over. So our answer is not “they are criminals” — it is **“nobody can put a price on this, therefore a fund looking after other people's savings has no business owning it.”**